GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Electronic Devices (EDC) — Official GATE ECE Syllabus

Paper Code: EC • Section 3: Electronic Devices (9 - 11 Marks)

Weightage: 9 - 11 Marks • **Subject Code:** EDC • **Difficulty:** Physics & Conceptual Heavy

OFFICIAL GATE EXAMINATION PATTERN & MARKING SCHEME
• **Question Types:** Multiple Choice Questions (MCQ, 1/3 negative marks), Multiple Select Questions (MSQ, zero negative marks), Numerical Answer Type (NAT, zero negative marks).
• **Weightage:** General Aptitude (15 Marks) + Engineering Mathematics (13 Marks) + Core Technical Subjects (72 Marks) = **100 Total Marks (65 Questions, 180 Minutes)**.

SECTION: SEMICONDUCTOR PHYSICS & CARRIER TRANSPORT

- Energy bands in intrinsic and extrinsic semiconductors: Conduction band, valence band, bandgap energy (Eg for Si = 1.12 eV, Ge = 0.66 eV, GaAs = 1.42 eV).
- Direct vs. indirect bandgap semiconductors, electron and hole effective masses.
- Equilibrium carrier concentrations: Fermi-Dirac distribution function, Fermi level position, intrinsic carrier concentration (ni).
- Mass Action Law: n * p = ni^2 (valid in thermal equilibrium for non-degenerate semiconductors).
- Doping and carrier density: Donor and acceptor doping, charge neutrality condition.
- Carrier transport mechanisms: Drift current, carrier mobility, temperature dependence of mobility (lattice scattering vs. ionized impurity scattering), velocity saturation.
- Diffusion current, Fick's first and second laws, Einstein relation (D/mu = Vt = kT/q).
- Generation and recombination of carriers: Band-to-band, Shockley-Read-Hall (SRH), and Auger recombination. Minority carrier lifetime and diffusion length (L = sqrt(D * tau)).
- Continuity equation and Poisson's equation for semiconductor device modeling. Hall Effect and Hall coefficient (sign of majority carriers, carrier concentration, mobility determination).

SECTION: P-N JUNCTIONS, ZENER & SPECIAL DIODES

- P-N junction in thermal equilibrium: Built-in potential (Vbi = Vt * ln(Na*Nd / ni^2)), energy band diagram across junction.
- Depletion region under forward and reverse bias: Space charge density, electric field distribution (triangular for step junction, maximum at metallurgical junction), depletion width (W proportional to sqrt(Vbi - V)).
- Junction capacitances: Depletion (transition) capacitance (dominant in reverse bias) and Diffusion capacitance (dominant in forward bias due to stored minority carriers).
- Diode current equation (Shockley equation): I = I0 * (exp(V / eta*Vt) - 1). Ideality factor (eta = 1 for diffusion, eta = 2 for recombination).
- Breakdown mechanisms: Zener breakdown (tunneling in heavily doped junctions, Vz < 4V, negative temperature coefficient) vs. Avalanche breakdown (impact ionization in lightly doped junctions, Vz > 6V, positive temperature coefficient).
- Special diodes: Zener diode, Tunnel diode (negative differential resistance, quantum mechanical tunneling), Varactor diode, PIN diode.
- Optoelectronic devices: Light Emitting Diode (LED), Semiconductor Laser, Photodiode (PIN and Avalanche photodiode, quantum efficiency, responsivity), Solar cell (fill factor, open circuit voltage, short circuit current).

SECTION: BIPOLAR JUNCTION TRANSISTOR (BJT)

- BJT physical structure and operating principles: NPN and PNP doping profiles (Emitter heavily doped, Base narrow and lightly doped, Collector large area for heat dissipation).
- Modes of operation: Forward active, Saturation, Cutoff, and Reverse active modes.

- Current components in active BJT: Base transport factor (beta*), Emitter injection efficiency (gamma), common base current gain (alpha), common emitter current gain (beta = alpha / (1 - alpha)).
- Ebers-Moll model of the BJT and reciprocal relations.
- Base width modulation (Early effect): Effective base width narrowing, Early voltage (VA), output resistance (ro = VA / IC).
- Transit time effects and cutoff frequencies (fT and f_max).

SECTION: MOS CAPACITOR & MOSFET PHYSICS

- MOS capacitor physics: Metal-Oxide-Semiconductor energy band diagram at equilibrium, work function difference (Phi_ms).
- Three regimes of MOS capacitor operation: Accumulation, Depletion, and Inversion (weak and strong inversion).
- Flat-band voltage (Vfb), Threshold voltage (VT) derivation: Surface potential (phi_s = 2 * phi_F at onset of strong inversion), oxide capacitance (Cox = epsilon_ox / tox).
- High-frequency vs. low-frequency C-V characteristics of MOS capacitor (deep depletion under fast ramp).
- MOSFET I-V characteristics: Cutoff regime, Linear (triode) regime (ID = mu*Cox*(W/L)*((VGS - VT)*VDS - VDS^2/2)), Saturation (pinch-off) regime (ID = 0.5*mu*Cox*(W/L)*(VGS - VT)^2).
- Channel length modulation: Parameter lambda, output conductance, effective channel length.
- Subthreshold conduction, subthreshold swing (S ~ 60 mV/decade at room temperature).
- Body effect (substrate bias effect): Increase in threshold voltage with reverse substrate bias (gamma body effect parameter).
- Short-channel effects (SCE): Velocity saturation, drain-induced barrier lowering (DIBL), hot carrier injection, gate oxide leakage.

SECTION: INTEGRATED CIRCUIT (IC) FABRICATION BASICS

- Crystal growth (Czochralski method) and silicon wafer preparation.
- Oxidation: Dry oxidation (thin, high-density gate oxide) vs. Wet oxidation (thick field oxide, faster growth, Deal-Grove model).
- Photolithography: Photoresists (positive and negative), photomasks, resolution limits.
- Diffusion and Ion Implantation: Projected range (Rp), straggle (delta_Rp), junction depth control.
- Thin-film deposition: Chemical Vapor Deposition (CVD), Physical Vapor Deposition (sputtering and evaporation) for metallization.
- Etching techniques: Wet chemical isotropic etching vs. Dry anisotropic Reactive Ion Etching (RIE).
- CMOS fabrication process flow: Twin-well (twin-tub) technology, shallow trench isolation (STI), latch-up prevention.

RECOMMENDED STANDARD TEXTBOOKS & REFERENCES

- Semiconductor Physics and Devices — Donald A. Neamen (McGraw Hill)
- Physics of Semiconductor Devices — S.M. Sze & Kwok K. Ng (Wiley)
- Solid State Electronic Devices — Ben G. Streetman & Sanjay Banerjee (Pearson)

HIGH-YIELD GATE EXAM STRATEGY & REVISION TIPS

- Built-in potential Vbi depends log-wise on doping and decreases by ~2 mV/°C temperature increase.
- Maximum electric field in abrupt step junction: E_max = 2*Vbi / W = q*Na*xp / epsilon.
- In MOS capacitor C-V curves, at high frequency the inversion layer cannot respond to the AC signal, so capacitance remains at C_min.